

Testing Industrial Policymakers: Evidence from Brazil's BNDES

First-Stage Results

March 2026

Is the sector-level allocation of BNDES lending across municipalities GDP-optimal?

- BNDES disburses $\sim 3\%$ of GDP annually in subsidized industrial lending
- Existing literature (Lane 2025, Juhász 2018) shows IP “works” for targeted sectors
- **But:** does the policymaker target the *right* sectors?
- We derive and test the **first-order condition for optimality**: at an interior optimum, marginal reallocations across sectors have zero effect on GDP

Contribution: Move from “does IP work?” to “is IP *optimal*?” — and if not, identify which sectors are over/under-supported

Theory: Local Optimality Test

Sector shares $s_m \in \Delta^{J-1}$, municipal output $Y_m = Y_m(s_m)$.

Directional derivative condition — at any interior optimum:

$$\left. \frac{dY_m}{d\varepsilon} \right|_{\varepsilon=0} = \nabla_{s_m} Y_m \cdot v_m = 0 \quad \text{for all } v_m \perp \mathbf{1}$$

Empirical projection (within-simplex, $\mathbf{1}'\Delta s_{mt} = 0$):

$$\Delta Y_{m,t+h} = \beta' \Delta s_{mt} + u_{m,t+h}$$

Null hypothesis (optimality):

$$H_0 : \quad \beta = \mathbf{0}$$

- Model-free: valid under any smooth objective
- $\hat{\beta}_j > 0$: sector j under-supported $\implies$ tilt more toward j
- $\hat{\beta}_j < 0$: sector j over-supported

Identification: Shift-Share IV from Political Alignment I

Δs_{mt} is endogenous $\implies$ need instruments that shift the sector mix exogenously.

Shift — Political alignment turnover:

- When party p wins office $\ell \in \{\text{mayor, governor, president}\}$, BNDES credit tilts toward p -affiliated firms (Lazzarini et al. 2015, Carvalho 2014)
- $\Delta \text{Align}_{mpt}^{\ell}$: party- p alignment shock in municipality m , year t

Share — Baseline sectoral exposure to each party:

$$w_{jmp,\tau} \equiv \frac{\sum_{f \in \mathcal{F}(j,m)} L_{f,p,\tau}}{\sum_{f \in \mathcal{F}(j,m)} L_{f,\tau}}$$

where L_{fp} = number of owners of firm f affiliated to party p ; L_f = total owners (affiliated and unaffiliated).

Identification: Shift-Share IV from Political Alignment II

Instrument (sector-specific, within municipality):

$$Z_{jmt}^{\ell} \equiv \sum_p w_{jmp,\tau} \cdot \Delta \text{Align}_{mpt}^{\ell}$$

Estimation Strategy

Firm first stage (firm $\times$ municipality $\times$ year):

$$\mathbf{1}(\text{BNDES}_{fmt} > 0) = \sum_{\ell} \lambda_{\ell} \text{FA}_{fmt}^{\ell} + \gamma_f + \alpha_{mt} + u_{fmt}$$

- $\text{FA}_{fmt}^{\ell} = \sum_p \omega_{fp,\tau} \text{Align}_{mpt}^{\ell}$: firm-level alignment exposure
- FE: firm + municipality $\times$ year; clustering: firm + municipality; weights: employment
- Also estimated for intensive margin ($\log \text{BNDES} \mid \text{BNDES} > 0$) and in changes

Sector first stage (municipality $\times$ sector $\times$ year):

$$\Delta s_{jmt} = \sum_{\ell} \lambda_{\ell} Z_{jmt}^{\ell} + \phi \text{EC}_{jm,\tau} + \gamma_{jm} + \alpha_{jt} + u_{jmt}$$

- $\text{EC}_{jm,\tau}$: exposure control (overall political connectedness of sector-municipality cell)
- Primary FE: municipality $\times$ sector + sector $\times$ year; clustering: municipality + sector
- Identification from **cross-sector variation** in party exposure $w_{jmp,\tau}$

Data Sources and Sector Groups

Source	Content	Coverage
BNDES portal	Loan-level	2002–2024
RAIS (MTE)	Employer-employee	2002–2017
TSE / Colonnelli	Party affiliation	2002–2019
IBGE PIB Muni	Municipality GDP	2002–2017
IBGE Population	Population	1991–2025

21 CNAE 2.0 sections → 10 sector groups:

- Manufacturing split into Light (CL), Heavy (CH), Advanced (CA)
- Utilities + Construction merged (UCo)
- Services: Trade (Tr), Transport (Tp), Market Services (MS), Public/Social (PSO)
- Residual (XX) dropped from estimation

Geographic unit: 5,570 municipalities **Sample:** 2002–2017

► sector distribution

Firm First Stage: Levels, Extensive Margin (Coalition)

	Dep. var: 1(BNDES _{fmt} > 0)					
	Mayor	Governor	President	M+G	M+P	All
$FA_{coal.}^{mayor}$	0.010** (0.005)			0.009* (0.005)	0.011** (0.005)	0.010** (0.005)
$FA_{coal.}^{gov}$		0.007 (0.005)		0.006 (0.005)		0.008 (0.005)
$FA_{coal.}^{pres}$			-0.015*** (0.006)		-0.016*** (0.006)	-0.018*** (0.006)
F-statistic	4.2	1.9	6.8	2.4	6.5	4.4
Observations	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589
R ²	0.450	0.450	0.450	0.450	0.450	0.450

Firm + muni × year FE. Coalition, cycle-specific. SEs clustered by firm + muni in parentheses. Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm First Stage: Levels, Intensive Margin (Coalition)

	Dep. var: $\log(\text{BNDES}_{fmt})$					
	Mayor	Governor	President	M+G	M+P	All
$FA_{coal.}^{mayor}$	-0.002 (0.102)			-0.014 (0.106)	-0.007 (0.102)	-0.017 (0.106)
$FA_{coal.}^{gov}$		0.155 (0.146)		0.156 (0.150)		0.144 (0.152)
$FA_{coal.}^{pres}$			0.141 (0.128)		0.142 (0.128)	0.111 (0.128)
F -statistic	0.0	1.1	1.2	0.6	0.6	0.7
Observations	407,953	407,953	407,953	407,953	407,953	407,953
R^2	0.752	0.752	0.752	0.752	0.752	0.752

Firm + muni $\times$ year FE. Coalition, cycle-specific. Sample: $\text{BNDES} > 0$. SEs clustered by firm + muni in parentheses. Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm First Stage: Changes, Extensive Margin (Coalition)

	Dep. var: $\Delta 1(\text{BNDES}_{fmt} > 0)$					
	Mayor	Governor	President	M+G	M+P	All
$\Delta FA_{\text{coal.}}^{\text{mayor}}$	0.005* (0.003)			0.005* (0.003)	0.005 (0.003)	0.005 (0.003)
$\Delta FA_{\text{coal.}}^{\text{gov}}$		0.001 (0.003)		0.001 (0.003)		0.000 (0.003)
$\Delta FA_{\text{coal.}}^{\text{pres}}$			-0.009*** (0.003)		-0.009*** (0.003)	-0.009*** (0.003)
F-statistic	2.7	0.1	9.0	1.4	5.3	3.6
Observations	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045
R^2	0.065	0.065	0.065	0.065	0.065	0.065

Firm + muni $\times$ year FE. Coalition, cycle-specific. SEs clustered by firm + muni in parentheses. Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm First Stage: Changes, Intensive Margin (Coalition)

	Dep. var: $\Delta \log(\text{BNDES}_{fmt})$					
	Mayor	Governor	President	M+G	M+P	All
$\Delta FA_{coal.}^{mayor}$	-0.060 (0.154)			-0.058 (0.152)	-0.066 (0.153)	-0.064 (0.151)
$\Delta FA_{coal.}^{gov}$		0.464** (0.230)		0.464** (0.230)		0.451* (0.232)
$\Delta FA_{coal.}^{pres}$			-0.287** (0.140)		-0.289** (0.141)	-0.268** (0.136)
<i>F</i> -statistic	0.2	4.1	4.2	2.0	2.1	2.4
Observations	128,675	128,675	128,675	128,675	128,675	128,675
R^2	0.467	0.467	0.467	0.467	0.467	0.467

Firm + muni $\times$ year FE. Coalition, cycle-specific. Sample: $\text{BNDES} > 0$ in both t and $t - 1$. SEs clustered by firm + muni in parentheses. Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm First Stage: Levels, Extensive Margin (Party)

	Dep. var: 1($\text{BNDES}_{fmt} > 0$)					
	Mayor	Governor	President	M+G	M+P	All
$FA_{\text{party}}^{\text{mayor}}$	0.021* (0.011)			0.021* (0.011)	0.021* (0.011)	0.021* (0.011)
$FA_{\text{party}}^{\text{gov}}$		0.000 (0.007)		-0.001 (0.007)		-0.001 (0.007)
$FA_{\text{party}}^{\text{pres}}$			-0.003 (0.013)		-0.005 (0.013)	-0.005 (0.013)
F-statistic	3.7	0.0	0.1	2.0	1.8	1.3
Observations	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589
R^2	0.450	0.450	0.450	0.450	0.450	0.450

Firm + muni $\times$ year FE. Party alignment, cycle-specific. SEs clustered by firm + muni in parentheses.

Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm First Stage: Changes, Extensive Margin (Party)

	Dep. var: $\Delta 1(\text{BNDES}_{fmt} > 0)$					
	Mayor	Governor	President	M+G	M+P	All
$\Delta FA_{party}^{\text{mayor}}$	0.003 (0.006)			0.003 (0.006)	0.003 (0.006)	0.003 (0.006)
$\Delta FA_{party}^{\text{gov}}$		-0.008 (0.005)		-0.008 (0.005)		-0.008 (0.005)
$\Delta FA_{party}^{\text{pres}}$			0.001 (0.005)		0.001 (0.005)	-0.000 (0.005)
F-statistic	0.3	2.6	0.0	1.3	0.2	0.9
Observations	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045
R^2	0.065	0.065	0.065	0.065	0.065	0.065

Firm + muni $\times$ year FE. Party alignment, cycle-specific. SEs clustered by firm + muni in parentheses.

Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector First Stage: Baseline (Coalition, Sector×Year FE)

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.001 (0.002)			-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
$Z_{coal.}^{gov}$		-0.003 (0.003)		-0.003 (0.003)		-0.003 (0.003)
$Z_{coal.}^{pres}$			0.001 (0.002)		0.001 (0.002)	0.000 (0.002)
F -statistic	0.2	0.6	0.1	1.3	0.3	0.9
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition alignment, cycle-specific weights, with municipality-sector exposure controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector First Stage: Muni×Year FE (Coalition)

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.000 (0.002)			-0.001 (0.002)	-0.000 (0.002)	-0.001 (0.002)
$Z_{coal.}^{gov}$		-0.003 (0.002)		-0.003 (0.002)		-0.003 (0.002)
$Z_{coal.}^{pres}$			-0.000 (0.002)		-0.000 (0.002)	-0.001 (0.002)
F -statistic	0.1	1.8	0.0	1.5	0.0	1.0
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.060	0.060	0.060	0.060	0.060	0.060

Muni × sector + muni × year FE. Coalition, cycle-specific, with municipality-sector exposure control interactions where identified. SEs clustered by muni + sector in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector First Stage: Levels (Coalition, s_{jmt} on Z_{jmt}^{levels})

	Dep. var: s_{jmt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{\text{coal.}}^{\text{mayor,lev}}$	-0.004 (0.002)			-0.004 (0.003)	-0.004 (0.003)	-0.004 (0.003)
$Z_{\text{coal.}}^{\text{gov,lev}}$		0.003 (0.005)		0.004 (0.005)		0.003 (0.005)
$Z_{\text{coal.}}^{\text{pres,lev}}$			0.005 (0.006)		0.005 (0.006)	0.004 (0.006)
F -statistic	3.0	0.6	0.5	1.6	1.5	1.1
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.350	0.350	0.350	0.351	0.351	0.351

Muni $\times$ sector + sector $\times$ year FE. Levels: alignment level $\times$ exposure weights. Coalition, cycle-specific, with municipality-sector exposure controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector First Stage: Party Alignment

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
Z_{party}^{mayor}	-0.000 (0.003)			-0.000 (0.003)	-0.000 (0.003)	-0.000 (0.003)
Z_{party}^{gov}		-0.003 (0.006)		-0.003 (0.006)		-0.003 (0.006)
Z_{party}^{pres}			0.004 (0.004)		0.004 (0.004)	0.003 (0.003)
F -statistic	0.0	0.2	1.0	0.4	0.9	0.8
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Party alignment, cycle-specific, with municipality-sector exposure controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sector First Stage: Without Exposure Control

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.001 (0.003)			-0.001 (0.002)	-0.001 (0.003)	-0.001 (0.002)
$Z_{coal.}^{gov}$		-0.002 (0.003)		-0.002 (0.003)		-0.002 (0.003)
$Z_{coal.}^{pres}$			0.001 (0.002)		0.001 (0.002)	0.000 (0.002)
F -statistic	0.2	0.4	0.2	1.3	0.2	0.9
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition, cycle-specific, no municipality-sector exposure control interactions. SEs clustered by muni + sector in parentheses. *** $p < 0.01$,

** $p < 0.05$, * $p < 0.10$.

Key Takeaways

Firm first stage:

- Political alignment predicts firm-level BNDES access and loan size
- Effect operates through both extensive and intensive margins
- Validates the micro political-lending channel underlying the sector-level instruments

Sector first stage:

- Baseline (sector $\times$ year FE): instruments are weak under coalition alignment
- Municipality $\times$ year FE and year-only FE yield stronger first stages
- Party-level alignment produces distinct variation at subnational tiers
- Exposure control matters: coefficients drop without it
- 2002-fixed weights are weaker than cycle-specific across the board

Main question: which FE structure and alignment definition should be primary? Coalition with sector $\times$ year FE is the most demanding but weakest specification.

Appendix: Firm — 2002-Fixed Baseline, Levels

	Dep. var: 1($\text{BNDES}_{fmt} > 0$)					
	Mayor	Governor	President	M+G	M+P	All
$FA^{\text{mayor}}_{\text{coal.}}$	0.014 (0.009)			0.014* (0.009)	0.016* (0.009)	0.016* (0.009)
$FA^{\text{gov}}_{\text{coal.}}$		-0.002 (0.010)		-0.003 (0.009)		-0.001 (0.009)
$FA^{\text{pres}}_{\text{coal.}}$			-0.031*** (0.009)		-0.032*** (0.009)	-0.032*** (0.010)
F-statistic	2.4	0.0	10.5	2.6	6.8	5.0
Observations	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589
R^2	0.450	0.450	0.450	0.450	0.450	0.450

Firm + muni $\times$ year FE. Coalition, 2002-fixed weights. SEs clustered by firm + muni in parentheses.

Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Firm — 2002-Fixed Baseline, Changes

	Dep. var: $\Delta 1(\text{BNDES}_{fmt} > 0)$					
	Mayor	Governor	President	M+G	M+P	All
$\Delta FA_{\text{coal.}}^{\text{mayor}}$	0.006 (0.004)			0.006 (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta FA_{\text{coal.}}^{\text{gov}}$		0.000 (0.004)		0.000 (0.004)		-0.001 (0.004)
$\Delta FA_{\text{coal.}}^{\text{pres}}$			-0.008** (0.003)		-0.008** (0.003)	-0.008** (0.003)
F-statistic	2.1	0.0	6.1	1.0	3.6	2.4
Observations	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045	32,881,045
R^2	0.065	0.065	0.065	0.065	0.065	0.065

Firm + muni $\times$ year FE. Coalition, 2002-fixed weights. SEs clustered by firm + muni in parentheses.

Weighted by employment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Firm — Unweighted, Levels

	Dep. var: 1($\text{BNDES}_{fmt} > 0$)					
	Mayor	Governor	President	M+G	M+P	All
$FA_{\text{coal.}}^{\text{mayor}}$	0.002*** (0.000)			0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
$FA_{\text{coal.}}^{\text{gov}}$		0.002*** (0.000)		0.002*** (0.000)		0.002*** (0.000)
$FA_{\text{coal.}}^{\text{pres}}$			-0.000 (0.000)		-0.000 (0.000)	-0.001** (0.000)
<i>F</i> -statistic	39.9	52.1	0.0	39.6	20.2	28.2
Observations	43,184,686	43,184,686	43,184,686	43,184,686	43,184,686	43,184,686
R^2	0.286	0.286	0.286	0.286	0.286	0.286

Firm + muni × year FE. Coalition, cycle-specific. Unweighted. SEs clustered by firm + muni in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Firm — Unweighted, Changes

	Dep. var: $\Delta 1(\text{BNDES}_{fmt} > 0)$					
	Mayor	Governor	President	M+G	M+P	All
$\Delta FA_{\text{coal.}}^{\text{mayor}}$	0.000 (0.000)			0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta FA_{\text{coal.}}^{\text{gov}}$		-0.000 (0.000)		-0.000 (0.000)		-0.000* (0.000)
$\Delta FA_{\text{coal.}}^{\text{pres}}$			-0.001*** (0.000)		-0.001*** (0.000)	-0.002*** (0.000)
<i>F</i> -statistic	1.6	1.3	29.7	1.4	15.4	11.3
Observations	35,497,793	35,497,793	35,497,793	35,497,793	35,497,793	35,497,793
R^2	0.025	0.025	0.025	0.025	0.025	0.025

Firm + muni × year FE. Coalition, cycle-specific. Unweighted. SEs clustered by firm + muni in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Sector — Year FE Only (Coalition)

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.001 (0.002)			-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
$Z_{coal.}^{gov}$		-0.003 (0.003)		-0.003 (0.003)		-0.004 (0.003)
$Z_{coal.}^{pres}$			0.000 (0.001)		0.000 (0.002)	-0.000 (0.001)
<i>F</i> -statistic	0.1	1.0	0.0	2.1	0.1	1.4
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.008	0.008	0.008	0.008	0.008	0.008

Muni $\times$ sector + year FE. Coalition, cycle-specific, with municipality-sector exposure controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Sector — 2002-Fixed Weights

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.005 (0.004)			-0.005 (0.004)	-0.005 (0.004)	-0.005 (0.004)
$Z_{coal.}^{gov}$		-0.001 (0.003)		-0.001 (0.003)		-0.001 (0.003)
$Z_{coal.}^{pres}$			-0.000 (0.002)		-0.000 (0.002)	-0.000 (0.002)
<i>F</i> -statistic	1.3	0.1	0.0	1.3	1.8	2.2
Observations	745,740	745,740	745,740	745,740	745,740	745,740
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition, 2002-fixed weights, with municipality-sector exposure controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix: Sector Group Distribution

Group	Label	Mean s_{mjt}	% Positive	N obs
Tr	Trade	0.185	38.9%	89,136
Tp	Transport	0.126	30.8%	88,640
CL	Light Manufacturing	0.084	24.6%	81,056
CH	Heavy Manufacturing	0.056	20.5%	73,312
MS	Market Services	0.039	15.8%	89,088
Ag	Agriculture	0.036	11.1%	84,944
UCo	Utilities & Construction	0.033	14.9%	86,544
CA	Advanced Manufacturing	0.026	14.1%	64,992
Mi	Mining	0.023	9.9%	52,480
PSO	Public, Social & Other	0.006	6.5%	85,264
Total observations				795,456

Appendix: Sector Group Definitions

Group	Label	CNAE Sections
Ag	Agriculture	A
Mi	Mining	B
CL	Light Manufacturing	C (divisions 10–18: food, textiles, wood, paper, printing)
CH	Heavy Manufacturing	C (divisions 19–25: petroleum, chemicals, metals, rubber)
CA	Advanced Manufacturing	C (divisions 26–33: electronics, machinery, vehicles, equipment)
UCo	Utilities & Construction	D (electricity/gas), E (water/waste), F (construction)
Tr	Trade	G (wholesale and retail)
Tp	Transport	H (transport and storage)
MS	Market Services	I, J, L, M–N (accommodation, information, real estate, professional/admin)
PSO	Public, Social & Other	P (education), Q (health), R (arts), S (other services)